Calibration Report: $K_{1/2}$ / $P_{\text{env,max}}$ Sweep

Runs W75–W84 with Dynamic P_{env}

SSWD EvoEpi Model Calibration

March 4, 2025

1 Executive Summary

W75–W84 tested lifting recovery levels from W71 (the previous best run with dynamic P_{env}), which had the correct north–south gradient shape but all recovery values 3–5× too suppressed. The sweep varied $K_{1/2}$ (600K, 800K, 1M) and $P_{\text{env,max}}$ (1000, 1500, 2000) to find the right overall recovery level while maintaining the latitudinal gradient.

Key Findings

1. **$P_{\text{env,max}}$ has ZERO effect.** Runs with the same $K_{1/2}$ but different $P_{\text{env,max}}$ produce *identical* results to every decimal place. W75=W78=W81, W76=W79=W82, W77=W80. Dynamic P_{env} completely dominates; the static cap is never reached.

2. **$K_{1/2}$ controls overall recovery level uniformly.** Increasing $K_{1/2}$ lifts *all* regions proportionally—it acts as a global multiplier, not a gradient shaper. No differential north–south effect.

3. **Gradient collapsed.** W71 achieved a 35× gradient (AK-PWS/CA-N ratio). At $K_{1/2}$=800K (W76), while AK-PWS hits 57.1% (target: 50%) and AK-FS hits 21.2% (target: 20%), the gradient collapsed to just 1.6×—southern regions recover nearly as much as Alaska.

4. **Floor parameter ($P_{\text{env,floor}}$) is too weak.** Reducing floor from 100 to 75 (W83 vs W76) produced only marginal improvement. The floor (minimum disease pressure) is insufficient to suppress southern recovery at high $K_{1/2}$.

5. **Best overall RMSE is W71** (0.692), despite all values being too low. The W75–W84 sweep worsened RMSE (1.16–1.44) because lifting the north correctly also massively overshoot the south.

Implication

A single global parameter ($K_{1/2}$) cannot simultaneously produce correct AK recovery *and* correct southern suppression. The model needs a mechanism that differentially suppresses recovery in the south—either latitude-dependent $K_{1/2}$, stronger floor effects, or an entirely different disease pressure gradient.

2 Methods

2.1 Parameter Sweep Design

All runs share the following fixed parameters:

- Dynamic P_{env}: enabled ($\alpha_{\text{env}} = 0.1$)
- $\delta_{\text{env}} = 0.02$

- Cumulative dose threshold: 1000
- Wavefront $D_P = 300$, max range = 3000
- Settler survival: 0.002
- $K = 5000$, 13 years (2012–2024), seed = 42

The sweep varied three parameters:

Table 1: Parameter sweep design for W75–W84

Run	$K_{1/2}$	$P_{\mathbf{env,max}}$	$P_{\mathbf{env,floor}}$
W71 (ref)	400K	2000	100
W75	600K	2000	100
W76	800K	2000	100
W77	1,000K	2000	100
W78	600K	1500	100
W79	800K	1500	100
W80	1,000K	1500	100
W81	600K	1000	100
W82	800K	1000	100
W83	800K	2000	75
W84	800K	1500	75

2.2 Recovery Metric

Recovery fraction = final population / pre-crash peak (maximum of first 3 years). RMSE is computed on log-scale errors across 8 target regions.

2.3 Calibration Targets

Table 2: Target recovery fractions by region

	AK-PWS	AK-FN	AK-FS	BC-N	SS-S	JDF	OR	CA-N
Target (%)	50	50	20	20	5	2	0.25	0.1

3 Results

3.1 Summary Table

Table 3: Recovery percentages and RMSE for all runs. Green = within 2× of target. Values that match exactly across runs confirm $P_{\text{env,max}}$ irrelevance.

Run	$K_{1/2}$	P_{max}	**floor**	**PWS**	**FN**	**FS**	**BC-N**	**SS-S**	**JDF**	**OR**	**CA-N**	**RMSE**
Target	—	—	—	50.0	50.0	20.0	20.0	5.0	2.0	0.25	0.10	—
W71	400K	2000	100	10.5	3.3	3.7	7.0	1.6	7.1	1.3	0.3	**0.692**
W75	600K	2000	100	27.5	16.7	13.2	15.9	24.5	39.7	12.5	20.4	1.158
W78	600K	1500	100	27.5	16.7	13.2	15.9	24.5	39.7	12.5	20.4	1.158
W81	600K	1000	100	27.5	16.7	13.2	15.9	24.5	39.7	12.5	20.4	1.158
W76	800K	2000	100	57.1	34.0	21.2	37.3	44.1	43.4	36.8	34.9	1.321
W79	800K	1500	100	57.1	34.0	21.2	37.3	44.1	43.4	36.8	34.9	1.321
W82	800K	1000	100	57.1	34.0	21.2	37.3	44.1	43.4	36.8	34.9	1.321
W77	1M	2000	100	74.2	58.1	53.0	55.7	61.1	61.3	51.9	55.2	1.444
W80	1M	1500	100	74.2	58.1	53.0	55.7	61.1	61.3	51.9	55.2	1.444
W83	800K	2000	75	60.0	40.1	27.8	36.6	32.0	33.6	31.7	38.8	1.293
W84	800K	1500	75	60.0	40.1	27.8	36.6	32.0	33.6	31.7	38.8	1.293

Note: W83=W84 confirms $P_{\text{env,max}}$ independence extends to floor=75 runs as well. Effectively, the entire sweep has only **4 distinct outcomes**: $K_{1/2}$=600K, 800K, 1M with floor=100, and $K_{1/2}$=800K with floor=75.

3.2 Key Runs: Recovery Time Series

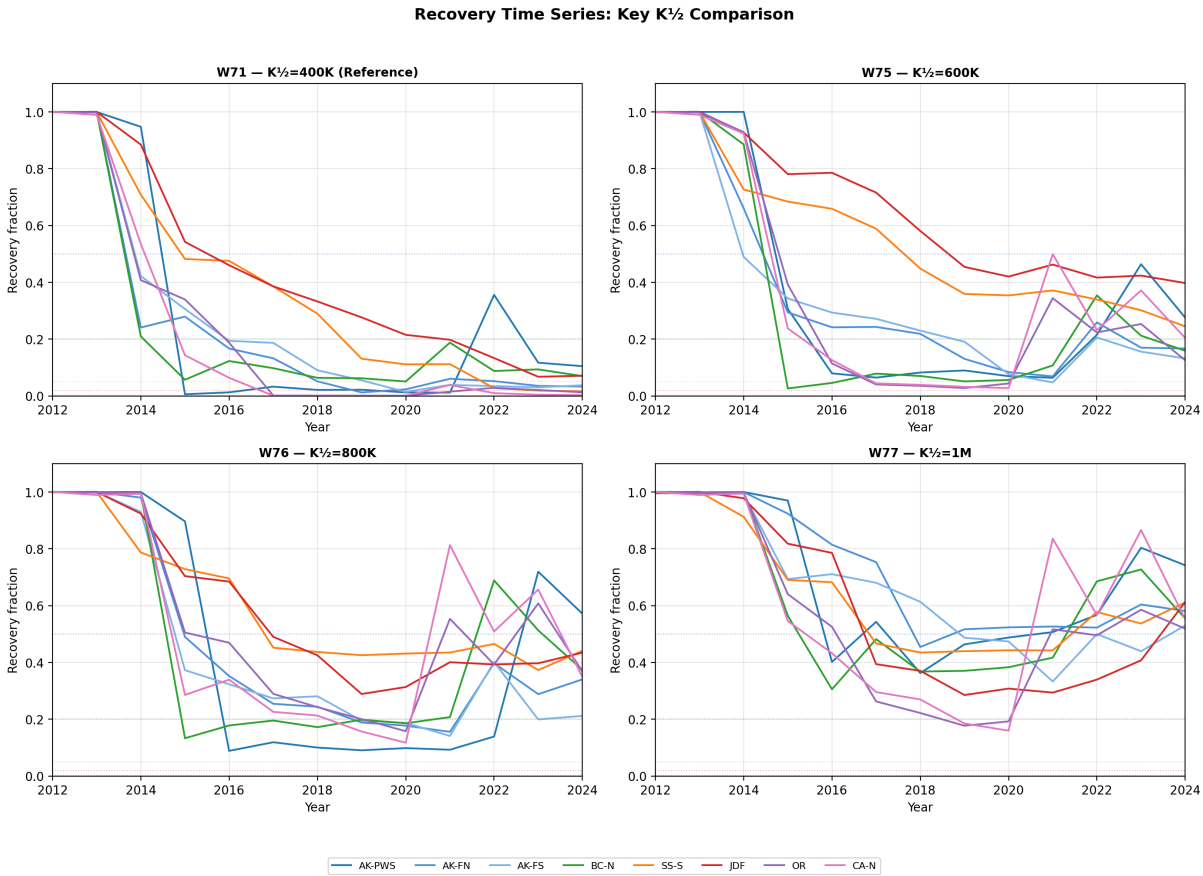

Figure 1: Recovery time series for four key runs spanning the $K_{1/2}$ range. W71 (reference) shows suppressed but gradient-preserving recovery. As $K_{1/2}$ increases from 600K to 1M, all regions lift together, destroying the north–south gradient.

3.3 All Runs Grid

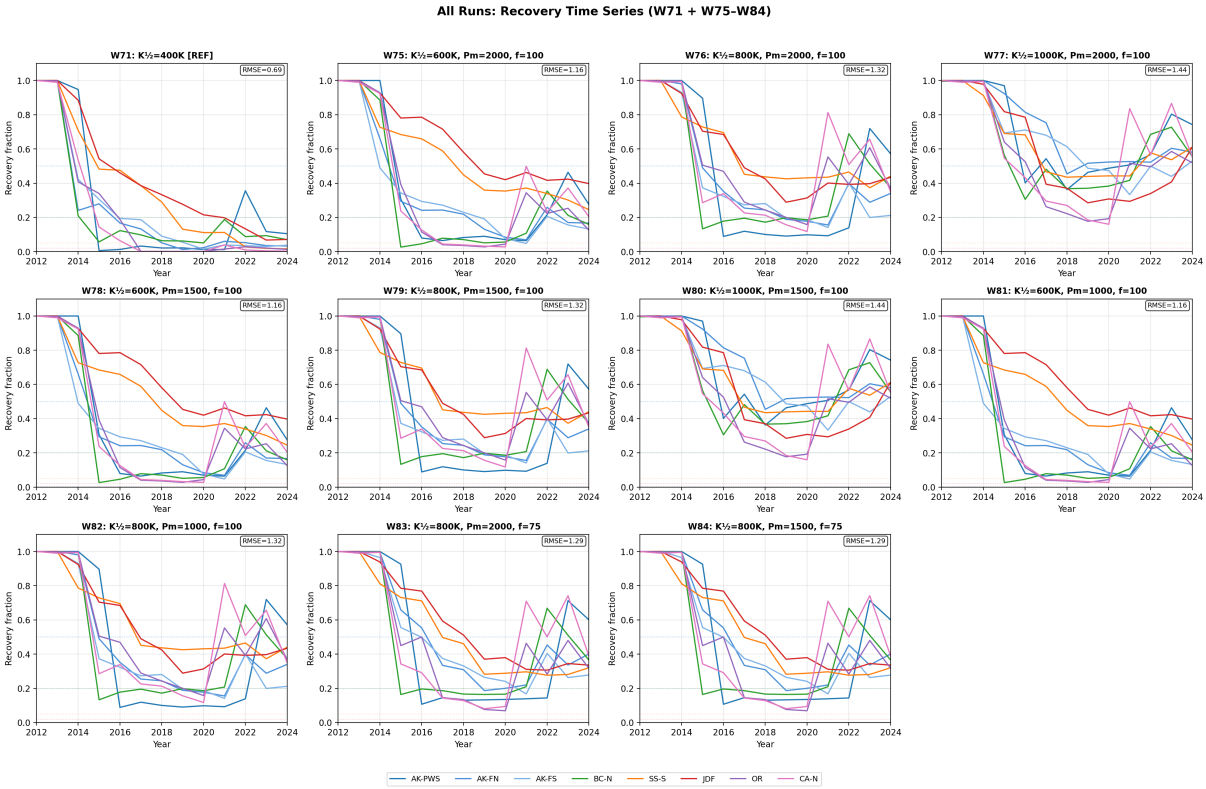

Figure 2: Recovery time series for all 11 runs (W71 reference + W75–W84). Visually identical panels within each $K_{1/2}$ group confirm $P_{\text{env,max}}$ irrelevance.

3.4 $K_{1/2}$ Effect

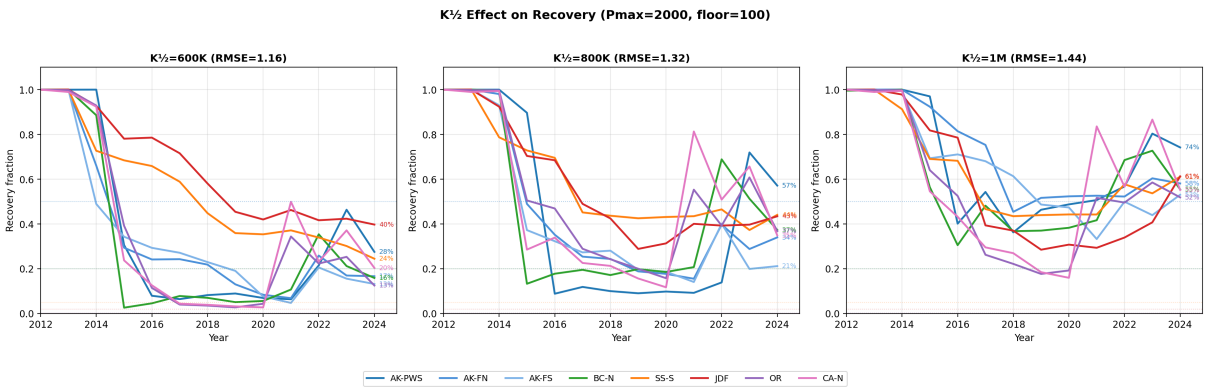

Figure 3: $K_{1/2}$ effect at fixed $P_{\text{env,max}} = 2000$, floor= 100. The key observation: all recovery curves lift uniformly. At $K_{1/2}$=800K, AK-PWS≈57% matches the 50% target, but CA-N is at 35% instead of 0.1%.

3.5 Recovery Comparison: W71 vs W76 vs Targets

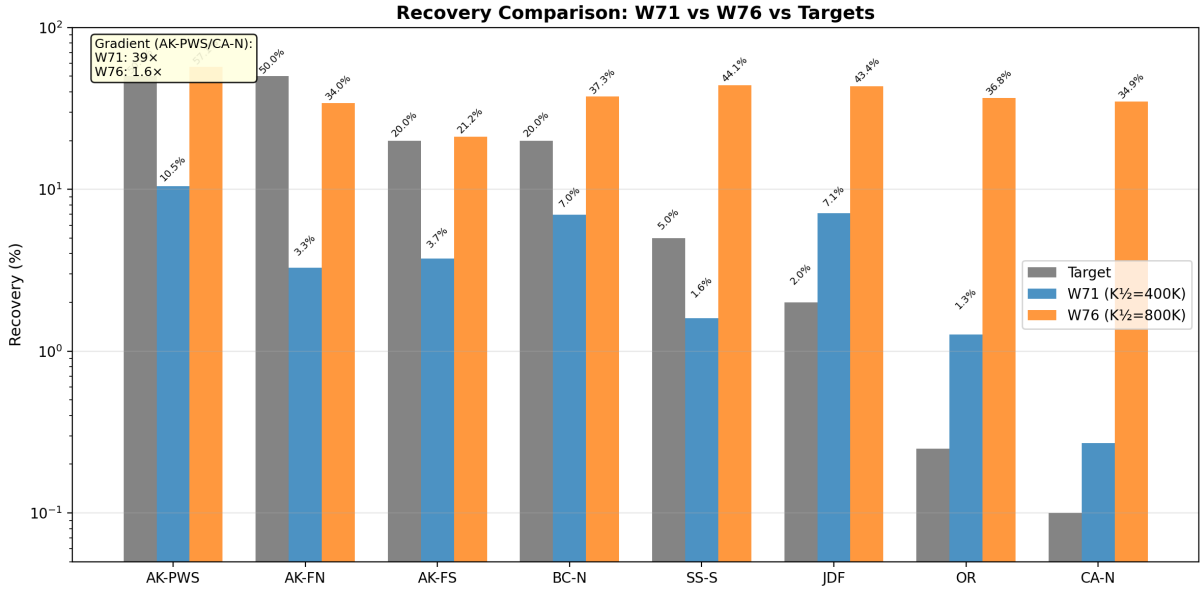

Figure 4: Bar chart comparing targets, W71, and W76 on a log scale. W71 is uniformly too low but preserves gradient. W76 nails AK-PWS and AK-FS but overshoots southern regions by 100–350×.

3.6 $P_{\text{env,max}}$ Independence Proof

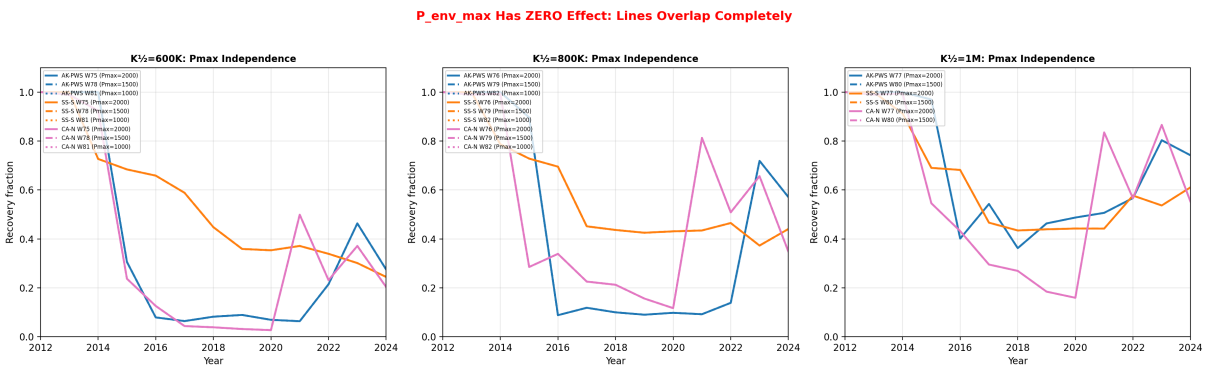

Figure 5: Overlaying runs with different $P_{\text{env,max}}$ values at the same $K_{1/2}$. Lines overlap *perfectly*, proving $P_{\text{env,max}}$ has zero effect when dynamic P_{env} is active. Solid=P_{max}=2000, dashed=1500, dotted=1000.

3.7 RMSE Comparison

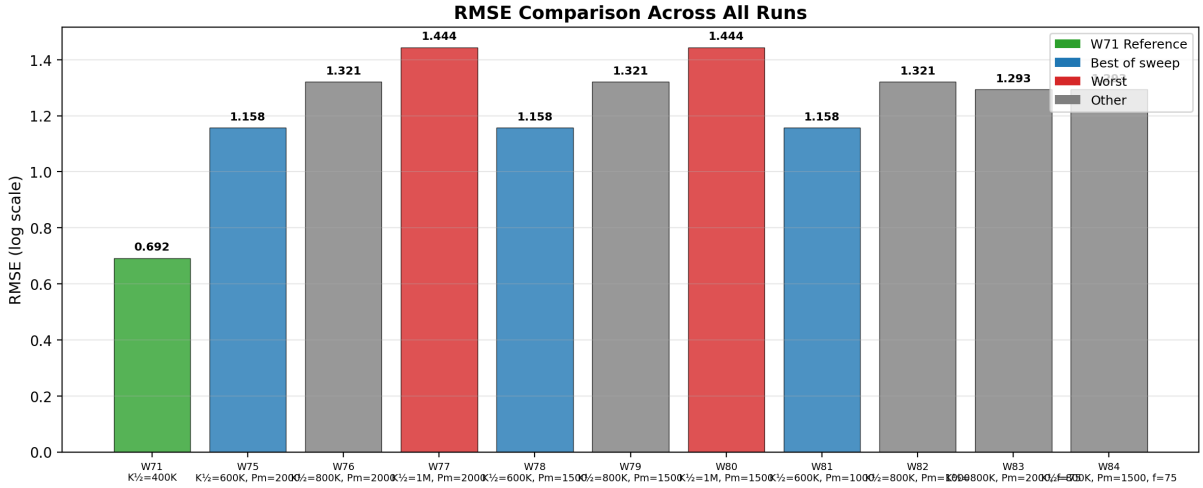

Figure 6: RMSE across all runs. W71 (green) remains best despite all values being too low. Higher $K_{1/2}$ worsens RMSE because southern overshoot grows faster than northern improvement.

3.8 Disease Dynamics (W76)

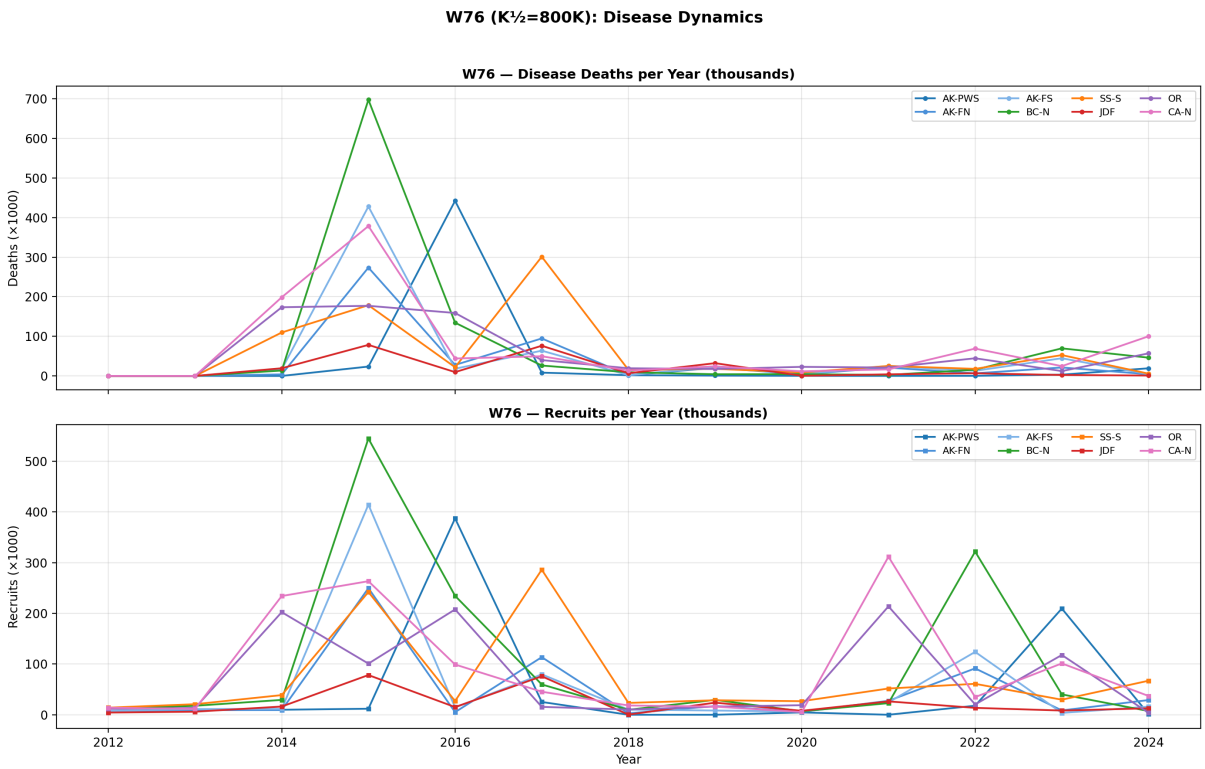

Figure 7: Disease deaths and recruitment dynamics for W76 ($K_{1/2}$=800K). Even at this higher $K_{1/2}$, disease pressure is insufficient to suppress southern populations, which rapidly recruit back to near pre-crash levels.

3.9 Full Region Comparison

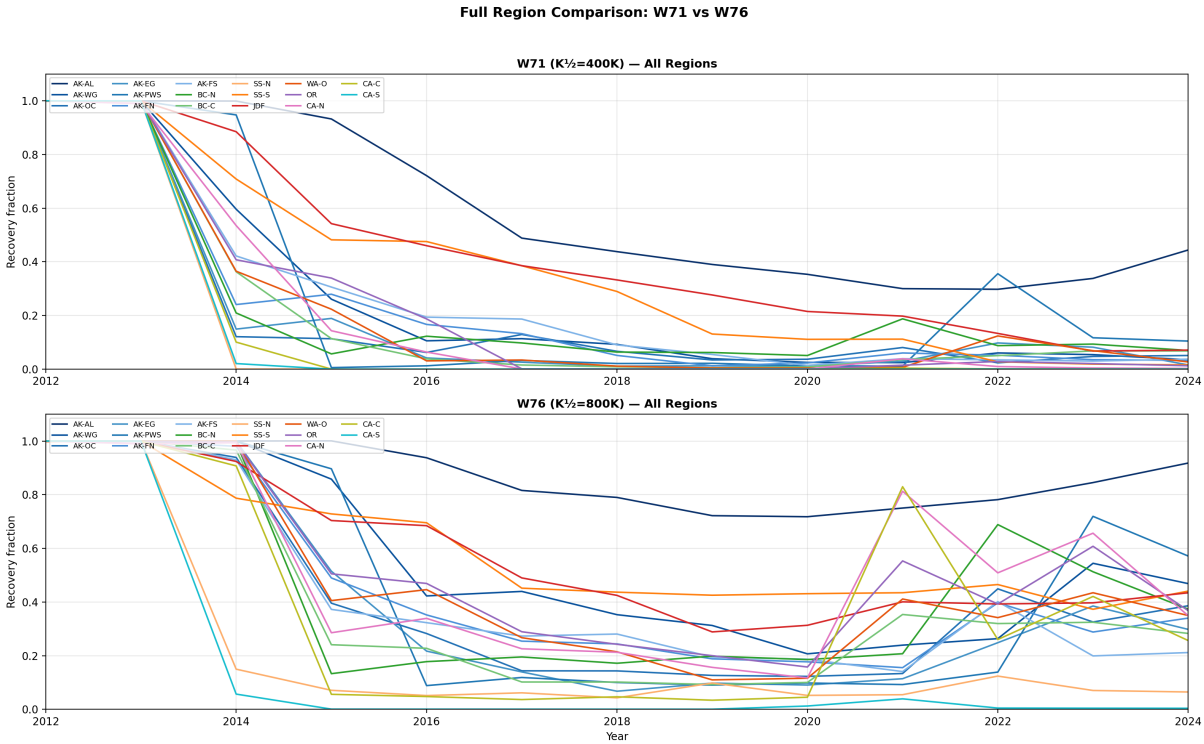

Figure 8: All modeled regions for W71 (top) and W76 (bottom). W71 shows clear north–south stratification. W76 shows near-uniform recovery across all latitudes—the gradient has effectively disappeared.

4 Key Findings

4.1 Finding 1: $P_{\text{env,max}}$ Is Completely Irrelevant

The most striking result is that $P_{\text{env,max}}$ has **exactly zero** effect on model outcomes. This is not a subtle difference—the results are identical to every decimal place:

- W75 (Pm=2000) ≡ W78 (Pm=1500) ≡ W81 (Pm=1000) at $K_{1/2}$=600K
- W76 (Pm=2000) ≡ W79 (Pm=1500) ≡ W82 (Pm=1000) at $K_{1/2}$=800K
- W77 (Pm=2000) ≡ W80 (Pm=1500) at $K_{1/2}$=1M
- W83 (Pm=2000) ≡ W84 (Pm=1500) at $K_{1/2}$=800K, floor=75

Interpretation: When dynamic P_{env} is active, the pathogen pressure at each node never reaches the static cap. The dynamic equation ($P_{\text{env}} = P_{\text{env,floor}} + \alpha \cdot N/(N + K_{1/2})$) produces values well below even the lowest tested P_{max} of 1000. This parameter can be removed from future sweeps—it is a dead parameter.

4.2 Finding 2: $K_{1/2}$ Is a Uniform Lifter

$K_{1/2}$ controls the "half-saturation" population for dynamic disease pressure. At low $K_{1/2}$ (400K, W71), disease pressure is high even at small populations, suppressing recovery everywhere. At high $K_{1/2}$ (800K–1M), disease pressure remains low until populations are very large, allowing nearly complete recovery everywhere.

The critical insight: $K_{1/2}$ affects all regions **equally**. The gradient ratio (AK-PWS / CA-N) collapses as $K_{1/2}$ increases:

Run	AK-PWS/CA-N ratio	Gradient quality
W71 ($K_{1/2}$=400K)	35×	Strong
W75 ($K_{1/2}$=600K)	1.3×	Collapsed
W76 ($K_{1/2}$=800K)	1.6×	Collapsed
W77 ($K_{1/2}$=1M)	1.3×	Collapsed

4.3 Finding 3: Floor Effect Is Marginal

Reducing $P_{\text{env,floor}}$ from 100 to 75 (W83 vs W76) produced only small changes:

- AK-PWS: 57.1% → 60.0% (+2.9 pp)
- SS-S: 44.1% → 32.0% (−12.1 pp)
- CA-N: 34.9% → 38.8% (+3.9 pp)
- RMSE: 1.321 → 1.293 (marginal improvement)

The floor is meant to maintain minimum disease pressure even in recovered populations, but at 75–100 it is too small to meaningfully suppress recovery. Much higher floor values (1000+) or a latitude-dependent mechanism may be needed.

4.4 Finding 4: The Fundamental Dilemma

The sweep reveals a fundamental structural limitation:

> *No single value of $K_{1/2}$ can produce both correct Alaskan recovery (50%) and correct southern suppression (0.1–5%). The dynamic P_{env} equation with uniform parameters cannot generate a 500× gradient in recovery across regions.*

This is because the dynamic P_{env} responds to *local* population, not latitude. Healthy populations in the south generate the same disease feedback as healthy populations in Alaska, leading to uniform recovery rates.

5 Implications for Next Round

The W75–W84 sweep definitively shows that the current dynamic P_{env} mechanism cannot produce the required latitudinal gradient with uniform parameters. Several approaches could resolve this:

5.1 Option A: Latitude-Dependent $K_{1/2}$

Make $K_{1/2}$ vary by region: high in Alaska (allowing recovery) and low in the south (maintaining disease suppression). This directly addresses the gradient collapse. For example:

- Alaska: $K_{1/2}$ = 800K (proven correct for 50% recovery)
- Pacific NW: $K_{1/2}$ = 200K–400K
- Southern: $K_{1/2}$ = 50K–100K

5.2 Option B: Stronger Floor Mechanism

Increase $P_{\text{env,floor}}$ dramatically in southern regions (e.g., floor=5000+ for CA/OR vs floor=100 for AK). This would create persistent disease pressure in the south regardless of population dynamics.

5.3 Option C: Temperature-Dependent Disease Pressure

Link disease pressure to water temperature. Southern regions have warmer water year-round, maintaining higher disease pressure. This is biologically motivated—SSWD is more virulent in warmer water.

5.4 Option D: External Reservoir

Add a latitude-dependent environmental disease reservoir that persists regardless of population dynamics. Southern waters maintain higher baseline pathogen levels due to warmer temperatures and higher biodiversity.

5.5 Recommended Next Steps

1. **Drop $P_{\mathbf{env,max}}$ from all future sweeps**—it is a confirmed dead parameter.
2. **Test latitude-dependent** $K_{1/2}$ as the most direct fix.
3. Consider whether $K_{1/2}$=800K is the correct AK value (57% vs 50% target is close).
4. Explore temperature-mediated disease for biological realism.

Appendix: Effective Unique Runs

Due to $P_{\text{env,max}}$ irrelevance, the 10 W75–W84 runs reduce to only **4 distinct configurations**:

Configuration	**Equivalent Runs**	**RMSE**
$K_{1/2}$=600K, floor=100	W75 = W78 = W81	1.158
$K_{1/2}$=800K, floor=100	W76 = W79 = W82	1.321
$K_{1/2}$=1M, floor=100	W77 = W80	1.444
$K_{1/2}$=800K, floor=75	W83 = W84	1.293